

TAIVALKOSKI KK URHEILUTIE, Finland

WMO: 028040

Lat: 65.575N

Lon: 28.218E

Elev: 197

StdP: 98.98

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.5	(c) -29.5	(d) -35.9	(e) 0.1	(f) -32.3	(g) -32.6	(h) 0.2	(i) -29.5	(j) 7.7	(k) -2.7	(l) 7.0	(m) -3.3	(n) 0.7	(o) 50	(p) 0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	26.0	16.8	23.7	15.9	21.7	15.3	18.7	23.2	17.5	21.9	16.3	20.2	3.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.6	20.0	15.7	11.4	19.1	14.5	10.5	18.0	54.0	23.5	49.9	21.8	46.3	20.4	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.2	6.3	5.6	-34.5	27.1	3.6	2.6	-37.1	29.0	-39.2	30.5	-41.2	32.0	-43.8	33.9
			-34.2	19.6	3.3	1.8	-36.6	20.9	-38.5	21.9	-40.3	22.9	-42.7	24.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.2	-12.4	-10.5	-7.0	-0.3	7.0	11.5	15.2	12.6	8.0	1.1	-3.6	-8.1
	DBStd	10.71	7.93	8.00	6.10	4.52	4.77	4.12	3.60	3.37	3.45	4.15	5.40	7.57
	HDD10.0	3593	696	573	527	309	118	30	4	13	78	275	407	562
	HDD18.3	6273	954	806	786	559	352	208	109	179	310	533	657	821
	CDD10.0	378	0	0	0	0	25	76	164	95	18	1	0	0
	CDD18.3	17	0	0	0	0	0	4	12	2	0	0	0	0
	CDH23.3	222	0	0	0	0	14	49	133	25	0	0	0	0
	CDH26.7	33	0	0	0	0	0	8	22	3	0	0	0	0
Wind	WSAvg	2.7	2.3	2.6	2.8	2.8	3.0	2.9	2.5	2.5	2.7	2.7	2.7	2.6
Precipitation	PrecAvg	689	48	41	38	33	55	71	94	74	62	60	58	55
	PrecMax	937	85	75	66	58	105	132	143	155	112	127	106	92
	PrecMin	550	15	9	14	9	12	27	40	30	7	12	16	17
	PrecStd	98	18	19	14	15	19	28	30	30	27	26	24	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	3.7	5.6	15.5	25.1	27.8	29.1	26.6	19.2	11.0	6.7	2.1
		MCWB	1.1	2.3	2.4	8.2	13.5	16.8	18.9	18.7	14.4	10.2	6.0	1.6
	2%	DB	0.1	1.8	3.6	10.5	22.8	24.5	26.8	23.0	16.6	8.8	5.0	1.3
		MCWB	-0.2	1.0	1.0	5.9	13.3	16.0	17.7	16.8	12.6	7.7	4.3	0.8
	5%	DB	-0.9	0.6	2.0	8.3	19.5	21.4	24.6	21.1	15.1	7.7	3.3	0.7
		MCWB	-1.1	0.1	0.3	3.8	12.0	14.2	16.9	15.7	11.8	6.8	2.9	0.4
	10%	DB	-2.4	-0.6	0.9	6.4	16.0	19.2	22.4	18.9	13.6	6.5	2.0	0.1
		MCWB	-2.8	-1.0	-0.2	2.8	9.8	13.1	16.1	14.6	11.4	5.7	1.7	-0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	2.4	3.0	8.9	15.4	18.7	20.6	19.6	15.2	10.4	6.4	1.7
		MCDB	1.6	3.2	5.0	13.7	21.7	23.2	24.4	23.3	18.1	10.9	6.7	1.7
	2%	WB	-0.1	1.1	1.6	6.0	13.7	17.2	19.2	17.8	13.5	8.2	4.5	1.0
		MCDB	0.1	1.8	2.8	9.8	21.2	23.0	23.6	21.5	15.5	8.6	4.8	1.1
	5%	WB	-1.1	0.1	0.7	4.3	12.2	15.6	18.2	16.4	12.5	6.9	2.9	0.4
		MCDB	-0.9	0.5	1.6	7.0	19.2	20.2	22.9	19.5	14.4	7.4	3.3	0.6
	10%	WB	-2.8	-1.0	-0.1	3.3	10.3	13.8	17.0	15.2	11.5	5.7	1.7	-0.2
		MCDB	-2.5	-0.6	0.8	5.6	15.0	17.8	21.3	18.1	13.2	6.5	1.9	0.0
Mean Daily Temperature Range	5% DB	MDBR	7.3	8.2	10.6	9.8	10.3	9.5	9.8	9.2	7.4	5.1	4.8	6.2
		MCDBR	5.7	5.7	8.6	12.8	16.7	13.9	14.0	13.1	10.5	6.1	4.3	4.2
		MCWBR	5.8	5.4	6.9	8.0	8.3	6.2	6.2	7.1	6.8	5.1	4.3	4.1
	5% WB	MCDBR	5.7	5.5	6.0	10.5	15.2	11.2	11.4	10.7	8.2	5.7	4.2	4.0
		MCWBR	5.8	5.2	5.1	6.8	7.9	5.6	5.7	6.3	5.7	5.2	4.2	4.0
Clear-Sky Solar Irradiance	taub	0.200	0.269	0.300	0.334	0.343	0.343	0.362	0.342	0.318	0.292	0.351	0.201	
	taud	1.930	2.102	2.150	2.231	2.449	2.475	2.440	2.477	2.528	2.365	2.648	1.958	
	Ebn at Noon	265	593	758	827	864	870	837	822	765	588	263	51	
	Edh at Noon	20	57	88	105	94	94	95	83	63	45	19	6	
All-Sky Solar Radiation	RadAvg	0.10	0.59	2.02	3.77	4.71	5.11	4.80	3.51	1.90	0.66	0.15	0.02	
	RadStd	0.01	0.12	0.19	0.37	0.45	0.39	0.48	0.35	0.20	0.10	0.03	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (31 neighbors)	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page